

Simriti Koul

+1 (703) 453 2852 - [Portfolio](#) - simriti.koul@yahoo.com - [LinkedIn](#) - [GitHub](#)

EDUCATION

George Mason University

Master of Science in Information Security and Assurance

Vellore Institute of Technology (VIT University)

Bachelor of Technology in Computer Science and Engineering

Fairfax, VA, USA

Aug 2021 - May 2023

Vellore, Tamil Nadu, India

July 2017 - June 2021

TECHNICAL SKILLS

Programming Languages: Python, Java, JavaScript, C++, HTML, CSS, R

Libraries and Tools: VS Code, SQL Developer, Linux, Agile, AWS, Power BI, VMware, Splunk, OpenCV, Office365, Autopsy, SaaS, Microsoft Office, Yara, Virus Total, MITRE, PyTorch, Apache Spark, PySpark, GCP, Git, Docker

SaaS: Wireshark, SailPoint, CyberArk, Microsoft Purview, Archer, Hexadecimal

Database: MySQL, eDiscovery

WORK EXPERIENCE

• Lecturer – Information Technology & Cybersecurity Program, Utah State University - Eastern (On-site), UT, USA, Aug 2025 – Present

- Delivering full-time instruction in Information Technology and Cybersecurity courses, blending theoretical knowledge with hands-on applications to support student learning and career readiness.
- Designing, preparing, and teaching undergraduate-level courses for the Fall 2025 and Spring 2026 semester, including:
 - TEIT-2200-QT1 – Security+: Covering core cybersecurity principles and preparing students for CompTIA Security+ certification.
 - BCIS-1500-PT1 – Introduction to Scripting: Introducing students to scripting fundamentals using Python for problem-solving and automation.
 - BCIS-1550-IO1 - Introduction to Python: Introducing students to scripting fundamentals using Python for problem-solving and automation.
 - TEIT-1550-GO1 XL - Practical Python: Introducing students to scripting fundamentals using Python for problem-solving and automation.
 - TEIT-1100-QT1 – Introduction to Networking: Teaching networking concepts, protocols, and troubleshooting skills aligned with industry standards.
 - ELET-2210-QT2 & ELET-2220-QL2 – Advanced PLC + Lab: Providing advanced instruction in programmable logic controllers (PLCs) with practical lab applications.
 - TEIT-2100-JO1 XL - Computer Networks: Designed to prepare students for supporting multi-platform networks and aligns with the CompTIA Network+ certification.
- Engaging in curriculum development, course material preparation, and assessment to enhance student outcomes.
- Mentoring students on academic progress, career pathways, and industry certifications.

• Data Input Specialist, Nomi Health, TX, USA, Oct 2024 – Present

- Developed Python scripts to automate data collection, perform impact analysis, and generate actionable insights from OPPS and IPPS data, improving efficiency for claims specialists.
- Built interactive dashboards and reports in DOMO and Power BI, translating complex datasets into clear visual narratives that supported data-driven decisions.
- Performed ETL operations using SQL and Snowflake to extract, transform, and load data, ensuring accuracy, consistency, and compliance with business requirements.
- Collaborated with cross-functional teams to define reporting needs, validate data accuracy, and deliver tailored visualizations aligned with strategic goals.
- Led data migration efforts and responded to critical data queries, resolving discrepancies and supporting business operations with timely, reliable insights.

• Software Engineer (Data), Blueicon Inc, VA, USA, May 2024 - Oct 2024

- Engineered and maintained CI/CD pipelines using Jenkins (master-slave architecture), Docker, and Git, automating build, test, and deployment processes across multiple environments to enhance delivery speed and reliability.

- Developed and deployed containerized applications with optimized Dockerfiles, integrating Bash scripting and Jenkinsfiles to automate infrastructure tasks and streamline operations in scalable microservice environments.
 - Led secure software development efforts, implementing secure coding practices to mitigate risks like SQL injection, XSS, and buffer overflows, and integrating security throughout the SDLC.
 - Conducted security assessments, including threat modeling, static/dynamic code analysis, and penetration testing, resulting in improved application resilience and compliance with security standards.
 - Collaborated across development, QA, and security teams, conducting code reviews, managing Git workflows, and delivering training to elevate secure coding awareness and foster cross-functional alignment.
- **Information Systems Auditor, North Carolina Agricultural & Technical State University, NC, USA, Feb 2024 - May 2024**
 - Conducted comprehensive information system audits using AuditBoard, ensuring internal controls were effective and compliant with policies and regulations, which enhanced operational efficiency and risk mitigation.
 - Reviewed and synthesized findings from studies, reports, and frameworks (NIST, FedRAMP, DFARS, COBIT, ISO 27002:2013), presenting clear conclusions and recommendations to leadership for informed decision-making.
 - Participated in the planning and execution of audit engagements, coordinating the collection of data across functional areas and ensuring smooth project delivery in alignment with compliance objectives.
 - Led internal control reviews and evaluations of computer information systems, applying risk management techniques and Agile methodologies to safeguard assets and ensure adherence to legal and institutional requirements.
 - Leveraged Advanced Excel, Tableau, and Power BI for detailed data analysis and visualization in financial, compliance, and investigative audits, significantly improving audit efficiency and investigative insights.
 - **Mobile Application Developer - Security, Resilience Inc, FL, USA, Sept 2023 - Feb 2024**
 - Developed and optimized MY SELENA and AIMEE apps using advanced JavaScript and SQL, increasing user engagement by 25% and reducing data access time by 30%, with a focus on Agile development and UX/UI optimization.
 - Led cross-functional teams through Agile and SDLC methodologies, cutting development cycles by 20% and enhancing team productivity and project management efficiency.
 - Pioneered new app features for the SEL curriculum using OpenCV and ETL processes, improving app functionality by 40% and boosting user satisfaction through data analytics and cloud computing with AWS.
 - Enhanced backend infrastructure for mobile apps, leveraging AWS and VMware to improve system scalability, reduce downtime by 35%, and strengthen app stability and security with cloud architecture and network administration.
 - Conducted detailed research and analysis of data, processes, and trends to inform app development, optimizing performance analytics and ensuring alignment with business goals and user needs.
 - **Student Research Assistant, George Mason University, VA, USA, Apr 2022 - May 2023**
 - Secured NSF project funding for an eye tracking experiment, leveraging expertise in research proposal development to investigate developmental disabilities, achieving a significant milestone in academic research and funding acquisition.
 - Developed an eye movement tracking program using Python and OpenCV, transforming real-time eye movements into actionable research data, enhancing the team's ability to analyze developmental disorders with advanced computer vision techniques.
 - Engineered an interactive visualization tool with Python's OpenCV, converting complex eye movement data into an intuitive image slider format, improving data interpretability and stakeholder engagement.
 - Utilized the Twitter API to extract and analyze over 60,000 tweets, applying advanced Excel and Python for sophisticated pattern recognition, categorizing content as scam or quality, and providing actionable insights for social media analysis.
 - Facilitated team collaboration by documenting code, methodologies, and processes, ensuring continuity and alignment across project milestones while contributing to the research on ADHD, Autism, and neuro-typical disorders.

PUBLICATIONS

- Simran Koul, Yash Raj, Simriti Koul. '**Analyzing Cyber Trends in Online Financial Frauds using Digital Forensics Techniques**', International Journal of Innovative Technology and Exploring Engineering (IJITEE) [Link](#)
- Yash Raj, Simriti Koul, Simran Koul. '**A Comparative Study of Different Image Encryption and Decryption Techniques**'. International Journal of Cyber-Security and Digital Forensics (IJCSDF) [Link](#)
- Simriti Koul, Udit Singhanian. '**Flower Species Detection over Large Dataset Using Convolution Neural Networks and Image Processing Techniques**'. International Journal of Innovative Science and Research Technology (IJISRT) [Link](#)
- Simriti Koul, '**Application of Deep Learning in Urban Sounds Classification**'. Turkish Online Journal of Qualitative Inquiry (TONQI) [Link](#)
- Prajval Mohan, Tejas Jambhale, Lakshya Sharma, Simran Koul, Simriti Koul. '**Load Balancing using Docker and Kubernetes: A Comparative Study**'. International Journal of Recent Technology and Engineering (IJRTE) [Link](#)
- Prajval Mohan, Adiksha Sood, Lakshya Sharma, Simran Koul, Simriti Koul. '**PC-SWT: A Hybrid Image Fusion Algorithm of Stationary Wavelet Transform and Principal Component Analysis**'. International Journal of Engineering and Advanced Technology (IJEAT) [Link](#)

PROJECTS

- Wordle Web Application Game, [GitHub](#)
- Optimizing Robot Navigation using SARSA and Q-Learning,
- Malware Analysis and Detection Investigation
- Collision Detection and Prevention System, [GitHub](#)
- Word Search Using Parallel Programming in Python, [GitHub](#)
- Complaint Register Portal, [GitHub](#)
- Location-Based Help Service, [GitHub](#)
- IoT Application on Wildlife Surveillance, [GitHub](#)
- Socket Programming, [GitHub](#)
- Connecting Devices and Message Communication using TCP/IP, [GitHub](#)
- Courier Information System, [GitHub](#)

CERTIFICATIONS

- **TCM SECURITY**: Practical Ethical Hacking The Complete Course